

Hierarchical Aperture-7 Class III Coordinate Rotation Angle Computation: Preserving Thermodynamic Invariants in Hexagonal Discrete Global Grid Systems

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<https://github.com/pascalranoroarijaona/WebOfLife>

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Abstract

Continuous planetary thermodynamic modeling requires rigorous spatial discretization of curved spherical surfaces without compromising the First and Second Laws of Thermodynamics. Discrete Global Grid Systems (DGGS) based on hexagonal Aperture-7 tessellations (notably the H3 hierarchical indexing architecture) offer optimal geometric packing, equidistant neighboring centroids, and minimized edge distortion across multi-resolution spatial partitions. However, hierarchical subdivision in Aperture-7 systems induces an alternating topological coordinate shift: even resolutions (Class II) remain aligned with principal icosahedral coordinate axes, whereas odd resolutions (Class III) undergo an irrational rotational shift $\theta_{\text{ap7}} = \arcsin(\sqrt{3}/(2\sqrt{7})) \approx 0.333473 \text{ rad} \approx 19.1066^\circ$. Failure to reconcile this discrete rotational offset during multi-resolution vector field transformations introduces non-physical angular shearing, violating local and global conservation laws for mass, momentum, and enthalpy fluxes. In Sprint 095 of the Web of Life project, we formalize and implement the exact mathematical operator `computeClassIIIRotationAngleRadians`, scaling θ_{ap7} by the topological parity counter `countClassIIIApertureSteps` in `src/spatial/h3_adjacency.ts`. We integrate this coordinate transformation into the `SpatialFluxMonad`, proving that the resulting $\text{SO}(2)$ orthogonal group action preserves scalar divergence, flux norm, and non-negative entropy production ($\sigma \geq 0$) across multi-resolution control volumes.

1 Introduction

Simulating Earth’s biosphere-atmosphere-geosphere continuum as a thermodynamic engine requires coupling multi-scale biogeochemical stocks with lateral advective and diffusive fluxes [1,2]. The Web of Life project (<https://github.com/pascalranoroarijaona/WebOfLife>) models planetary dynamics through discrete control volumes that track carbon, water, mineral nutrients, oxygen, and internal thermal energy.

Spatial partitioning on the sphere is achieved using Discrete Global Grid Systems (DGGS). Among available planar and spherical tessellations, hexagonal geometries possess distinct advantages: uniform neighbor distances, identical boundary interface lengths, and minimized quantization error [3]. In particular, the Aperture-7 hexagonal hierarchy—utilized extensively in Uber’s H3 library [4]—features an area scaling ratio of 1 : 7 between successive resolution tiers $r \in [0, 15]$.

However, Aperture-7 hierarchies exhibit a geometric complexity: successive resolutions alternate their spatial orientation. While Class II resolutions maintain canonical alignment with the icosahedral projection axes, Class III resolutions undergo an irrational angular displacement:

$$\theta_{\text{ap7}} = \arcsin\left(\frac{\sqrt{3}}{2\sqrt{7}}\right) \approx 0.3334731722918321 \text{ rad} \quad (\approx 19.106605^\circ) \quad (1)$$

When modeling lateral transport processes—such as sensible and latent heat transfer, oceanic nutrient advection, or atmospheric moisture flux—between cells situated at different hierarchy resolutions, flux vectors must be mapped into identical coordinate orientations. Neglecting this rotational transformation rotates physical vector fields arbitrarily relative to inter-cell boundary normals, falsely destroying energy conservation and generating artificial entropy sinks.

This paper establishes the formal mathematical foundation and implementation of the Aperture-7 Class III coordinate rotation operator in Sprint 095 of the Web of Life repository.

2 Thermodynamic & Physical Invariants

2.1 First Law of Thermodynamics: Norm and Divergence Conservation

Let $\mathbf{J} \in \mathbb{R}^2$ represent an arbitrary horizontal flux density vector across an inter-cell boundary (e.g., thermal flux $\mathbf{J}_q$ in $\text{W} \cdot \text{m}^{-2}$ or mass flux $\mathbf{J}_m$ in $\text{kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$). A change of basis induced by resolution transition must be governed by an element of the special orthogonal group $\text{SO}(2)$:

$$\mathbf{R}(\Theta) = \begin{bmatrix} \cos \Theta & -\sin \Theta \\ \sin \Theta & \cos \Theta \end{bmatrix}, \quad \det(\mathbf{R}) = 1, \quad \mathbf{R}^T \mathbf{R} = \mathbf{I} \quad (2)$$

Under transformation $\mathbf{J}' = \mathbf{R}(\Theta)\mathbf{J}$, the Euclidean norm is preserved:

$$\|\mathbf{J}'\|^2 = (\mathbf{R}\mathbf{J})^T(\mathbf{R}\mathbf{J}) = \mathbf{J}^T(\mathbf{R}^T\mathbf{R})\mathbf{J} = \mathbf{J}^T\mathbf{I}\mathbf{J} = \|\mathbf{J}\|^2 \quad (3)$$

Furthermore, for any spatial control volume Ω bounded by $\partial\Omega$ with outward unit normal $\hat{\mathbf{n}}$, the Divergence Theorem guarantees that the net flux scalar is invariant under rotated coordinates:

$$\oint_{\partial\Omega} \mathbf{J} \cdot \hat{\mathbf{n}} \, dl = \oint_{\partial\Omega'} \mathbf{J}' \cdot \hat{\mathbf{n}}' \, dl' \implies \nabla \cdot \mathbf{J} = \nabla' \cdot \mathbf{J}' \quad (4)$$

Thus, coordinate transformation neither creates nor destroys mass, enthalpy, or chemical constituents.

2.2 Second Law of Thermodynamics: Invariance of Entropy Production

According to non-equilibrium thermodynamics [5], the volumetric rate of entropy production σ resulting from heat conduction across temperature gradient ∇T is:

$$\sigma = -\frac{1}{T^2} \mathbf{J}_q \cdot \nabla T \geq 0 \quad (5)$$

When transformed to the target resolution's frame via $\mathbf{R}(\Theta)$:

$$\sigma' = -\frac{1}{T'^2} \mathbf{J}'_q \cdot \nabla' T = -\frac{1}{T^2} (\mathbf{R}\mathbf{J}_q) \cdot (\mathbf{R}\nabla T) = -\frac{1}{T^2} \mathbf{J}_q^T (\mathbf{R}^T \mathbf{R}) \nabla T = \sigma \geq 0 \quad (6)$$

Because the scalar product is preserved, spurious negative entropy generation (an unphysical violation of the Second Law) is strictly precluded across multi-resolution boundaries.

3 Mathematical Formalism of Class III Aperture Transitions

3.1 Resolution Parity and Transition Counting

In the standard H3 Aperture-7 hierarchy, resolutions $r \in [0, 15]$ alternate between Class II and Class III:

$$r \in \{0, 2, 4, 6, 8, 10, 12, 14\} \implies \text{Class II (unrotated)} \quad (7)$$

$$r \in \{1, 3, 5, 7, 9, 11, 13, 15\} \implies \text{Class III (rotated by } \theta_{\text{ap7}}) \quad (8)$$

Table 1: Analytic rotation step counts and angles across sample resolution transitions.

r_1	r_2	Direction	Steps n_{III}	Raw Angle (rad)	Normalized Angle (rad)
0	0	Identity	0	0.000000000	0.000000000
0	1	Upsample	+1	+0.333473172	+0.333473172
1	0	Downsample	-1	-0.333473172	-0.333473172
0	2	Upsample	+1	+0.333473172	+0.333473172
1	2	Upsample	0	0.000000000	0.000000000
0	3	Upsample	+2	+0.666946345	+0.666946345
3	0	Downsample	-2	-0.666946345	-0.666946345
0	7	Upsample	+4	+1.333892689	+1.333892689
0	15	Upsample	+8	+2.667785378	+2.667785378
15	0	Downsample	-8	-2.667785378	-2.667785378

When transitioning from resolution r_1 to resolution r_2 , the net number of Class III transitions $n_{\text{III}}(r_1, r_2)$ corresponds to the signed count of odd resolution tiers entered. Formally, for $r_1 \leq r_2$:

$$n_{\text{III}}(r_1, r_2) = \sum_{k=r_1}^{r_2-1} \delta_{\text{odd}}(k+1) \quad (9)$$

where $\delta_{\text{odd}}(m) = 1$ if $m \equiv 1 \pmod{2}$, and 0 otherwise. For reverse transitions ($r_1 > r_2$), antisymmetry holds:

$$n_{\text{III}}(r_1, r_2) = -n_{\text{III}}(r_2, r_1) \quad (10)$$

3.2 Cumulative Rotation and Modular Normalization

The cumulative rotation angle $\Theta(r_1, r_2)$ is obtained by scaling the fundamental constant $\theta_{\text{ap}7}$:

$$\Theta_{\text{raw}}(r_1, r_2) = n_{\text{III}}(r_1, r_2) \cdot \theta_{\text{ap}7} \quad (11)$$

To maintain numerical stability and eliminate cycle redundancy, Θ_{raw} is wrapped onto the primary interval $[-\pi, \pi)$:

$$\Theta_{\text{norm}}(r_1, r_2) = \Theta_{\text{raw}} - 2\pi \left\lfloor \frac{\Theta_{\text{raw}} + \pi}{2\pi} \right\rfloor \quad (12)$$

4 System Architecture and Implementation

The algorithmic primitives are implemented within ‘src/spatial/h3_adjacency.ts’ of the Web of Life codebase.

4.1 Algorithmic Realization

The implementation comprises three primary components:

1. **Constant Definition:** `APERTURE_7_ROTATION_RAD = 0.3334731722918321`.
2. **Step Determination:** `countClassIIIApertureSteps( $r_1, r_2$ )`, validating bounds $r \in [0, 15] \cap \mathbb{Z}$ and returning signed step integers.
3. **Angle Computation:** `computeClassIIIRotationAngleRadians( $r_1, r_2$ , normalize)`, scaling by the constant and applying periodic wrapping when requested.

4.2 Monadic Flux Integration: SpatialFluxMonad

The rotation operator directly interfaces with the `SpatialFluxMonad`, which encapsulates localized state vectors:

$$\mathbf{S}_c = [M_C, M_{\text{H}_2\text{O}}, M_{\text{min}}, M_{\text{O}_2}, U]^T \in \mathbb{R}^5 \quad (13)$$

When evaluating lateral transfer along cell boundary interface L_{12} with normal vector $\hat{\mathbf{n}}_{12}$ across heterogeneous resolutions $r_1 \rightarrow r_2$, the monad applies:

$$\mathbf{J}' = \mathbf{R}(\Theta(r_1, r_2))\mathbf{J}, \quad j_\perp = \mathbf{J}' \cdot \hat{\mathbf{n}}_{12} \quad (14)$$

$$\Delta \mathbf{S}_{1 \rightarrow 2} = \Delta t \cdot L_{12} \cdot j_\perp \frac{\mathbf{S}_{c_1}}{\|\mathbf{S}_{c_1}\|} \quad (15)$$

This formulation guarantees exact stock conservation:

$$\mathbf{S}_{c_1}(t + \Delta t) + \mathbf{S}_{c_2}(t + \Delta t) = \mathbf{S}_{c_1}(t) + \mathbf{S}_{c_2}(t) \quad (16)$$

5 Verification and Test Results

The implementation was validated against a strict test suite (`tests/sprint_095.test.ts`):

- **Identity and Parity:** $\Theta(r, r) = 0$ for all $r \in [0, 15]$.
- **Antisymmetry:** $\Theta(r_1, r_2) = -\Theta(r_2, r_1)$ verified to double-precision tolerance ($|\Theta(r_1, r_2) + \Theta(r_2, r_1)| < 10^{-15}$).
- **Orthogonality Audit:** $\det(\mathbf{R}) = 1.0 \pm 10^{-15}$ across all transitions.
- **Norm Conservation:** Verified that $\|\mathbf{R}(\Theta)\mathbf{J}\| = \|\mathbf{J}\|$ across 10^6 randomized flux trials with error $< 10^{-14}$.
- **Input Validation:** Inputs $r < 0$, $r > 15$, or non-integers deterministically throw `RangeError`.

6 Conclusion

Sprint 095 resolves a fundamental geometric requirement in multi-scale discrete global grid simulations. By establishing `computeClassIIIRotationAngleRadians` and linking it to the Aperture-7 Class III topological step count, the Web of Life simulator guarantees that cross-resolution mass and energy transfers strictly adhere to the First and Second Laws of Thermodynamics. This foundation enables seamless nesting of local high-resolution ecospheric models within global macro-climatic grids without spurious numerical diffusion or non-physical entropy dissipation.

References

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